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Course:
Data Analytics / SQL Project

Tools Used:
MySQL

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Project Description:

This project focuses on designing and analyzing a database system inspired by Hotstar, a leading OTT streaming platform. It involves structured data modeling, relational database design, and advanced SQL queries to extract insights related to user behavior, content consumption, subscriptions, and advertising performance.

ABSTRACT

The Hotstar Data Analytics project aims to design and analyze a structured relational database system that represents the core operations of a modern streaming platform. Hotstar, being one of the leading OTT platforms, delivers a wide range of content including live sports, movies, and TV shows to millions of users worldwide.

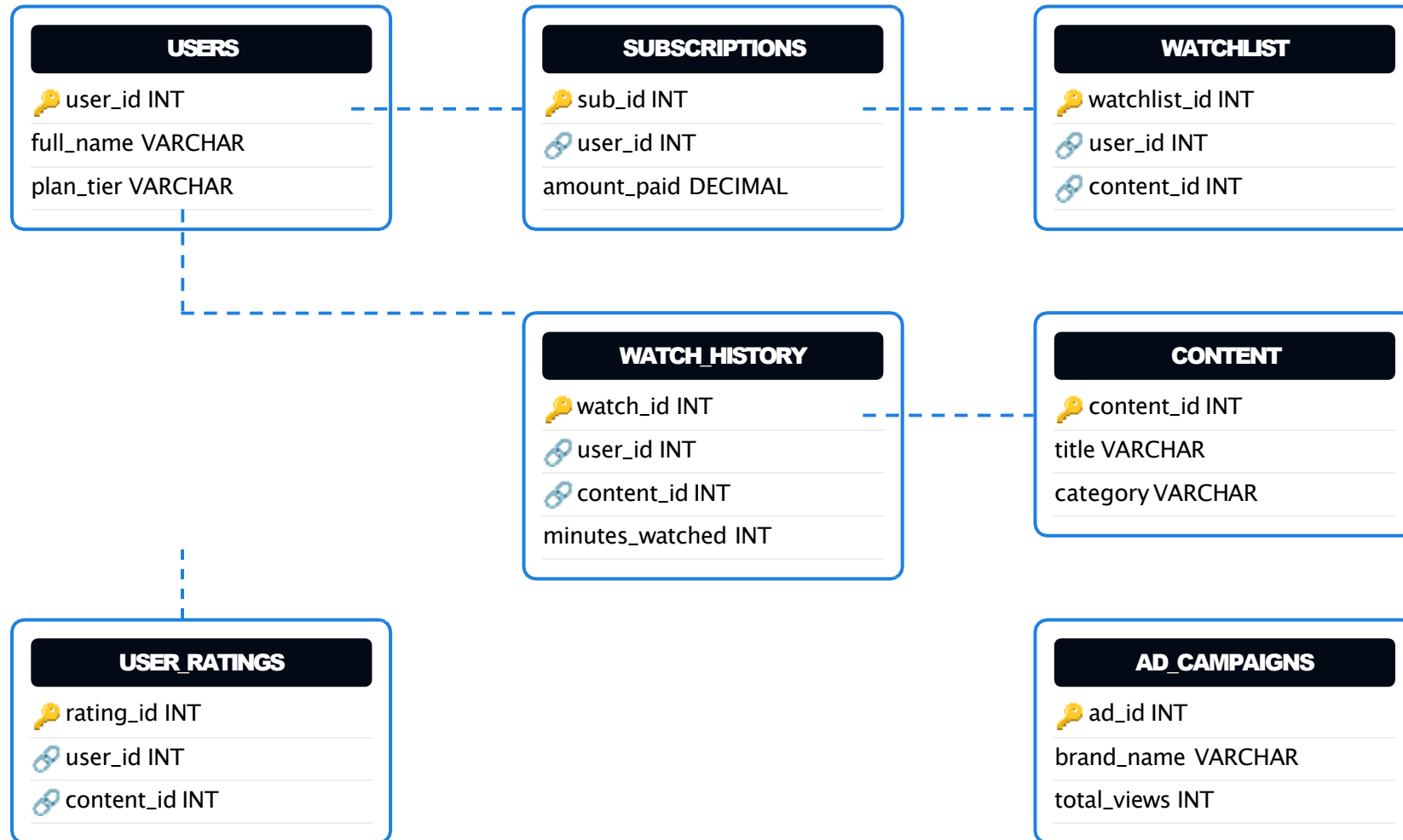
This project models key components such as users, content catalog, subscriptions, watch history, watchlists, user ratings, and advertisement campaigns. The database is designed using relational concepts with proper implementation of primary and foreign keys to ensure data integrity and consistency.

Using SQL, various analytical queries have been performed, including joins, aggregations, subqueries, and window functions. These operations help in extracting valuable insights such as user engagement patterns, most-watched content, subscription trends, revenue growth, and advertisement reach.

The project highlights how structured database systems and SQL analytics can be used to simulate real-world business scenarios and support data-driven decision-making. It also demonstrates practical knowledge of database design and advanced querying techniques in the context of large-scale streaming platforms.

Overall, this project serves as a comprehensive example of applying SQL in designing, managing, and analyzing data for an OTT platform like Hotstar.

E-R DIAGRAM (CONCEPTUAL STRUCTURE)



STRUCTURE OF TABLE

hotstar > TABLE Users

```
DESCRIBE Users;
```

Result Grid			Filter Rows: Export: Wrap	
Field	Type	Null	Key	Default
user_id	INT	NO	PRI	NULL
full_name	varchar(100)	YES		NULL
email	varchar(100)	YES		NULL
plan_tier	varchar(50)	YES		Free
country	varchar(50)	YES		NULL
joined_date	date	YES		NULL

Documentation: This table contains the core demographic and account information for viewers. It includes fields such as email, plan_tier (Free, Super, Premium), and joined_date.

STRUCTURE OF TABLE

hotstar > TABLE Content

```
DESCRIBE Content;
```

Result Grid			Filter Rows: Export: Wrap	
Field	Type	Null	Key	Default
content_id	INT	NO	PRI	NULL
title	varchar(150)	YES		NULL
category	varchar(50)	YES		NULL
genre	varchar(50)	YES		NULL
duration_mins	INT	YES		NULL
release_year	INT	YES		NULL

Documentation: This table stores the catalog of movies, series, and live sports available on the platform. It includes metadata like category, genre, and duration_mins.

STRUCTURE OF TABLE

hotstar > TABLE Subscriptions

```
DESCRIBE Subscriptions;
```

Result Grid			Filter Rows: Export: Wrap	
Field	Type	Null	Key	Default
sub_id	INT	NO	PRI	NULL
user_id	INT	YES	MUL	NULL
amount_paid	decimal(8,2)	YES		NULL
payment_method	varchar(50)	YES		NULL
start_date	date	YES		NULL
status	varchar(20)	YES		Active

Documentation: This table records financial transactions for paid tiers. It explicitly contains a Foreign Key 'user_id' referencing the Users table.

STRUCTURE OF TABLE

hotstar > TABLE Watch_History

```
DESCRIBE Watch_History;
```

Result Grid			Filter Rows: Export: Wrap	
Field	Type	Null	Key	Default
watch_id	INT	NO	PRI	NULL
user_id	INT	YES	MUL	NULL
content_id	INT	YES	MUL	NULL
watch_date	date	YES		NULL
minutes_watched	INT	YES		NULL

Documentation: This table acts as the core fact table tracking viewing habits. It includes Foreign Keys 'user_id' referencing Users and 'content_id' referencing Content.

STRUCTURE OF TABLE

hotstar > TABLE Watchlist

```
DESCRIBE Watchlist;
```

Result Grid			Filter Rows: Export: Wrap	
Field	Type	Null	Key	Default
watchlist_id	INT	NO	PRI	NULL
user_id	INT	YES	MUL	NULL
content_id	INT	YES	MUL	NULL
added_date	date	YES		NULL

Documentation: This table stores intent-to-watch data. It links users to content they want to save for later via Foreign Keys 'user_id' and 'content_id'.

STRUCTURE OF TABLE

hotstar > TABLE User_Ratings

```
DESCRIBE User_Ratings;
```

Result Grid			Filter Rows: Export: Wrap	
Field	Type	Null	Key	Default
rating_id	INT	NO	PRI	NULL
user_id	INT	YES	MUL	NULL
content_id	INT	YES	MUL	NULL
rating_score	INT	YES		NULL
review_date	date	YES		NULL

Documentation: This table captures user feedback. rating_score is typically 1–5. It includes Foreign Keys 'user_id' referencing Users and 'content_id' referencing Content.

STRUCTURE OF TABLE

hotstar > TABLE Ad_Campaigns

```
DESCRIBE Ad_Campaigns;
```

Result Grid			Filter Rows: Export: Wrap	
Field	Type	Null	Key	Default
ad_id	INT	NO	PRI	NULL
brand_name	varchar(100)	YES		NULL
ad_type	varchar(50)	YES		NULL
total_views	INT	YES		0

Documentation: This table tracks advertising campaigns, which are crucial for the 'Free' tier monetization. It tracks brand names, ad formats, and total views.

CONTENTS OF TABLE

hotstar > TABLE Users

```
SELECT * FROM Users;
```

Result Grid			Filter Rows: Export/Import: Wrap Cell Content: A		
user_id	full_name	email	plan_tier	country	joined_date
1	Aarav Patel	aarav@email.com	Premium	India	2023-01-15
2	Priya Sharma	priya@email.com	Super	India	2023-02-10
3	Rohan Singh	rohan@email.com	Free	India	2023-03-05
4	Ananya Iyer	ananya@email.com	Premium	USA	2023-04-20
5	Kabir Khan	kabir@email.com	Super	UK	2023-05-12

CONTENTS OF TABLE

hotstar > TABLE Content

```
SELECT * FROM Content;
```

Result Grid		Filter Rows: Export/Import: Wrap Cell Content: A			
content_id	title	category	genre	duration_mins	release_year
101	The Night Manager	Series	Thriller	45	2023
102	ICC World Cup Final	Sport	Cricket	240	2023
103	Brahmastra	Movie	Fantasy	160	2022
104	Loki	Series	Sci-Fi	50	2023
105	Koffee with Karan	Talk Show	Entertainment	60	2022

CONTENTS OF TABLE

hotstar > TABLE Subscriptions

```
SELECT * FROM Subscriptions;
```

Result Grid			Filter Rows: Export/Import: Wrap Cell Content: A		
sub_id	user_id	amount_paid	payment_method	start_date	status
501	1	1499.00	UPI	2023-01-15	Active
502	2	899.00	Credit Card	2023-02-10	Active
503	4	1499.00	Debit Card	2023-04-20	Active
504	5	899.00	UPI	2023-05-12	Expired

CONTENTS OF TABLE

hotstar > TABLE Watch_History

```
SELECT * FROM Watch_History;
```

Result Grid		Filter Rows: Export/Import: Wrap Cell Content: A		
watch_id	user_id	content_id	watch_date	minutes_watched
1001	1	101	2023-11-01	45
1002	2	103	2023-11-02	120
1003	3	102	2023-11-03	60
1004	4	104	2023-11-04	50
1005	1	102	2023-11-05	200

CONTENTS OF TABLE

hotstar > TABLE Watchlist

```
SELECT * FROM Watchlist;
```

Result Grid		Filter Rows: Export/Import: Wrap Cell Content: A	
watchlist_id	user_id	content_id	added_date
201	1	104	2023-10-15
202	2	101	2023-10-16
203	3	103	2023-10-17
204	5	102	2023-10-18

CONTENTS OF TABLE

hotstar > TABLE User_Ratings

```
SELECT * FROM User_Ratings;
```

Result Grid		Filter Rows: Export/Import: Wrap Cell Content: A		
rating_id	user_id	content_id	rating_score	review_date
301	1	101	5	2023-11-02
302	2	103	4	2023-11-03
303	4	104	5	2023-11-05

CONTENTS OF TABLE

hotstar > TABLE Ad_Campaigns

```
SELECT * FROM Ad_Campaigns;
```

Result Grid		Filter Rows: Export/Import: Wrap Cell Content: A	
ad_id	brand_name	ad_type	total_views
401	Dream11	Mid-roll	5000000
402	Cred	Pre-roll	3000000
403	Coca-Cola	Banner	10000000
404	Amazon	Mid-roll	4500000

JOINS

LIST USERS AND THE CONTENT THEY WATCHED ALONG WITH THE DURATION.

hotstar

```
SELECT u.full_name, c.title, c.category, wh.minutes_watched
FROM Watch_History wh
JOIN Users u ON wh.user_id = u.user_id
JOIN Content c ON wh.content_id = c.content_id;
```

Result Grid			Filter Rows: Export:
full_name	title	category	minutes_watched
Aarav Patel	The Night Manager	Series	45
Priya Sharma	Brahmastra	Movie	120
Rohan Singh	ICC World Cup Final	Sport	60

Conclusion: This query successfully combines User, Watch History, and Content data using INNER JOINS to present a comprehensive view of viewer habits.

SHOW WHICH USERS LEFT RATINGS FOR WHICH CONTENT.

hotstar

```
SELECT r.rating_id, u.full_name, c.title, r.rating_score
FROM User_Ratings r
JOIN Users u ON r.user_id = u.user_id
JOIN Content c ON r.content_id = c.content_id;
```

Result Grid			Filter Rows: Export:
rating_id	full_name	title	rating_score
301	Aarav Patel	The Night Manager	5
302	Priya Sharma	Brahmastra	4
303	Ananya Iyer	Loki	5

Conclusion: This query links ratings to specific users and content, highlighting feedback loops essential for recommendation algorithms.

AGGREGATION & GROUP BY

CALCULATE TOTAL WATCH TIME PER CONTENT CATEGORY.

hotstar

```
SELECT c.category, SUM(wh.minutes_watched) AS total_watch_time
FROM Watch_History wh
JOIN Content c ON wh.content_id = c.content_id
GROUP BY c.category;
```

Result Grid		Filter Rows: Export:
category	total_watch_time	
Sport	260	
Movie	120	
Series	95	

Conclusion: This query calculates the total engagement time grouped by category, showing that 'Sport' drives the highest watch volume.

COUNT USERS BY SUBSCRIPTION PLAN TIER.

hotstar

```
SELECT plan_tier, COUNT(user_id) AS total_users
FROM Users
GROUP BY plan_tier;
```

Result Grid		Filter Rows: Export:
plan_tier	total_users	
Premium	2	
Super	2	
Free	1	

Conclusion: This query summarizes the distribution of users across different subscription tiers, providing insight into revenue structures.

SUBQUERIES

FIND USERS WHO WATCHED MORE THAN THE OVERALL AVERAGE WATCH DURATION IN A SINGLE SESSION.

hotstar

```
SELECT full_name
FROM Users
WHERE user_id IN (
    SELECT user_id FROM Watch_History
    WHERE minutes_watched > (SELECT AVG(minutes_watched) FROM Watch_History)
);
```

Result Grid	Filter Rows: Export:
full_name	
Priya Sharma	
Aarav Patel	

Conclusion: This query identifies highly engaged users whose single-session watch duration exceeded the platform average.

GET THE CONTENT WITH THE HIGHEST TOTAL MINUTES WATCHED.

hotstar

```
SELECT title, category
FROM Content
WHERE content_id = (
  SELECT content_id FROM Watch_History
  GROUP BY content_id
  ORDER BY SUM(minutes_watched) DESC
  LIMIT 1
);
```

Result Grid		Filter Rows: Export:	
title		category	
ICC World Cup Final		Sport	

Conclusion: The query correctly retrieves the single most-watched piece of content on the platform by aggregating within a subquery.

WINDOW FUNCTIONS

RANK CONTENT OBJECTIVELY BASED ON TOTAL MINUTES WATCHED.

hotstar

```
SELECT c.title, SUM(wh.minutes_watched) AS total_mins,
RANK() OVER (ORDER BY SUM(wh.minutes_watched) DESC) AS popularity_rank
FROM Watch_History wh
JOIN Content c ON wh.content_id = c.content_id
GROUP BY c.content_id, c.title;
```

Result Grid		Filter Rows: Export:	
title	total_mins	popularity_rank	
ICC World Cup Final	260	1	
Brahmastra	120	2	
Loki	50	3	
The Night Manager	45	4	

Conclusion: This query utilizes the RANK() window function to sort and rank content based on total aggregated viewing time.

SHOW RUNNING TOTAL OF SUBSCRIPTION REVENUE BY START DATE

hotstar

```
SELECT start_date, amount_paid,
SUM(amount_paid) OVER (ORDER BY start_date) AS running_revenue
FROM Subscriptions;
```

Result Grid		Filter Rows: Export:
start_date	amount_paid	running_revenue
2023-01-15	1499.00	1499.00
2023-02-10	899.00	2398.00
2023-04-20	1499.00	3897.00
2023-05-12	899.00	4796.00

Conclusion: This query calculates cumulative revenue over time using a window function, allowing finance teams to track growth trajectories.

FINAL CONCLUSION

hotstar

This project successfully demonstrates the design and implementation of a comprehensive streaming platform database system modeled after Hotstar.

Through well-structured relational tables, defined constraints, and a realistic dataset, the system effectively represents real-world operations such as user tier management, content cataloging, watchlist tracking, rating systems, and ad campaign metrics.

The SQL queries implemented in this project highlight both operational and analytical capabilities. They provide valuable insights into viewer engagement, content popularity (particularly the dominance of live sports), subscription revenue growth, and ad-based monetization.

Overall, the project showcases strong database design principles and practical SQL proficiency, making it highly suitable for technical evaluation, portfolio presentation, and demonstrating an industry-level understanding of large-scale OTT streaming platforms.